Feel the Pain: An Interpretable Multimodal Approach for Physiological Signal-Based Pain Detection

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Abstract

Pain assessment is essential in clinical settings, but remains largely subjective, relying on verbal self-reports that are impractical for non-communicative patients. In this study, we propose an interpretable multimodal approach for automatic pain detection using physiological signals. Leveraging 65 participants (23 female, 42 male) multimodal data from the AI4Pain 2025 challenge, we systematically explore the contributions of multiple sensor modalities, Electrodermal Activity (EDA), Blood Volume Pulse (BVP), Respiration (RESP), and Peripheral Oxygen Saturation (SpO_2) to classify pain into **No Pain**, **Low Pain**, and **High Pain** categories. After careful pre-processing and feature extraction, XGBoost outperformed other machine learning and deep learning models. In particular, the combination of only EDA and BVP features achieved 80% validation accuracy. In addition, the model using combined EDA and BVP features achieved high classification accuracy across all pain levels, particularly for High Pain (AUC = 0.92) and No Pain (AUC = 0.96), demonstrating strong discriminative ability. Using all modalities, the XGBoost model achieved a test accuracy of 60.92%, marking an 8.62% improvement over the multimodal test baseline. Notably, our model outperformed the baseline results across all modalities on both the validation and test sets. Our model also incorporates HRV features derived from BVP and EDA emphasizes interpretability through analysis of permutation feature importance. The results demonstrate that multimodal physiological data, when processed effectively, enable accurate and explainable pain detection, a step toward real-time, deployable pain monitoring systems in healthcare.

CCS Concepts

• **Human-centered computing → Ubiquitous and mobile computing**.

Keywords

Pain Recognition, Physiology, Machine Learning

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1 Introduction

Pain assessment is one of the most critical and challenging tasks encountered by clinicians [17]. This becomes particularly critical

when clinicians need to assess pain in patients who cannot communicate (ie non-verbal patients). Although pain is a subjective experience that traditionally relies on patient self-reporting, the inability to express pain does not eliminate the possibility of its presence, nor the necessity for timely and effective pain management [18]. As a critical physiological and psychological experience, pain plays a fundamental role in clinical diagnosis and patient care, making accurate assessment essential for effective treatment planning, postoperative monitoring, and improving overall patient outcomes.

Despite its importance, current clinical practices rely heavily on self-reported pain scores, such as the Visual Analogue Scale (VAS) or the Numeric Rating Scale (NRS) [8]. These subjective methods are prone to bias, miscommunication, and are not feasible in scenarios where patients are also unconscious, cognitively impaired, or non-verbal (e.g., infants or individuals with dementia). Thus, developing an objective and reliable system for automatic pain recognition remains a significant challenge. Pain is a complex and multifaceted phenomenon influenced by numerous factors, including individual tolerance to pain, psychological state, and contextual signals.

Physiological signals offer a promising, non-invasive alternative for continuous pain monitoring. Signals such as Electrodermal Activity (EDA), Blood Volume Pulse (BVP), Respiration (RESP), and Peripheral Oxygen Saturation (SpO_2) reflect the autonomic nervous system's response to pain stimuli [5, 11, 23]. These signals can be acquired through wearable devices and provide real-time insight into the internal state of the body. When processed and analyzed appropriately, they can serve as valuable indicators for automated pain detection systems.

In this study, we propose a multimodal framework for pain level classification using physiological data provided in the AI4Pain 2025 challenge, jointly organized by the University of Oxford and the University of Canberra. The dataset includes multimodal recordings from multiple subjects under induced *No Pain*, *Low Pain*, and *High Pain* conditions. We performed a comprehensive analysis to identify the most effective combination of physiological signal modalities for classifying varying levels of pain. We defined baseline recordings as representing the *No Pain* condition and deliberately excluded post-stimulus rest periods to avoid the influence of lingering pain responses. To enable effective machine learning, we applied modality-specific preprocessing techniques, including band-pass filtering and signal decomposition, to enhance signal quality and extract meaningful physiological patterns.

Furthermore, to distinguish between *No Pain*, *Low Pain*, and *High Pain* states, we evaluated a range of traditional machine learning models, including Random Forest (RF), Gradient Boosting (GB), Support Vector Machine (SVM), and Extreme Gradient Boosting (XGBoost). Among various machine learning and deep learning models, XGBoost achieved the highest accuracy. For single modality, we obtained the highest validation accuracy with the EDA signal (73%), while on the test set, BVP achieved the highest accuracy of 62.93%, marking a 17.24% improvement over the test baseline when using BVP alone. We also investigated the best fusion of sensor modalities in terms of accuracy and found that combining EDA and BVP yielded 80% validation accuracy and 62.07% test accuracy. However, when we added respiration data to EDA and BVP, the validation accuracy remained the same (80%), and including other modalities with BVP did not improve the results.

To enhance model transparency, we applied permutation feature importance analysis. For EDA, tonic signal dynamics and skin conductance response rates were key predictors. For BVP, HRV metrics, especially inter-beat intervals and non-linear features contributed most to pain classification while EDA's most important features are derived from tonic/phasic decomposition and wavelet-based analysis. These findings align with known physiological pain responses, adding interpretability to our model. This work bridges the gap between raw physiological signal processing and effective feature extraction across multiple modalities, enabling robust models for objective and deployable pain recognition. Our contributions in this paper are threefold:

- We systematically evaluate the effectiveness of multiple physiological signals (EDA, BVP, RESP, SpO_2 for three-class pain classification.
- We demonstrate that a model combining EDA and BVP features achieves high accuracy, outperforming single-modality approaches and established baselines.
- We provide model interpretability through permutation feature importance, identifying key physiological markers of pain and enhancing clinical trustworthiness.

2 Related Work

Accurate and objective pain assessment has gained considerable attention in recent years, especially with the rise of machine learning and the increasing availability of physiological signal data [5, 7, 11, 19, 21, 23]. Traditional pain assessment methods, such as self-report scales (e.g., VAS, NRS) [8], although widely used, are subjective and unsuitable for patients unable to communicate. This limitation has driven research toward automated systems for pain recognition based on physiological signals.

Pain has a strong association with physiological responses, particularly through its interaction with the autonomic nervous system (ANS), which activates protective sympathetic mechanisms. Among the most common ANS-mediated indicators of pain are skin conductance responses (SCR) and heart-related metrics such as heart rate (HR) and heart rate variability (HRV) [4, 10, 15]. Skin conductance, which reflects electrodermal activity (EDA), is directly influenced by the sympathetic nervous system, as sweat glands are exclusively innervated by sympathetic excitatory fibers [3]. When pain occurs, increased sympathetic activity triggers sweat secretion, which alters the skin's electrical conductance until the sweat is either reabsorbed or evaporates [1, 2]. Prior work by Posada-Quintero et al. demonstrated that EDA features can be used to predict both the objective intensity of noxious stimuli and subjective pain perception, highlighting the strong potential of EDA as a physiological marker of pain.

2.1 Physiological Signals for Pain Detection

Numerous studies have employed machine learning techniques for pain detection [7, 11, 19, 21]. Fernandez Rojas et al. proposed a multimodal physiological sensing framework to assess acute pain by utilizing electrodermal activity (EDA), photoplethysmography (PPG), and respiration (RESP) signals collected from 22 participants exposed to electrical pain stimuli on the forearm and hand. A total

of 122 statistical and sensor-specific features were extracted and reduced using joint mutual information (JMI), and traditional models such as SVM, decision trees, and LDA were evaluated using leave-one-subject-out cross-validation. Their SVM-based model achieved 93.2% accuracy for binary classification, identifying EDA as the most informative modality [5]. Another recent systematic review further emphasized the importance of multimodal physiological sensing for acute pain assessment, highlighting the role of sensor fusion and AI methods in improving model robustness. The review analyzed 31 studies from 2015 to 2024, noting the growing use of cardiac, electrodermal, and neural signals in pain recognition and the increasing application of deep learning techniques [13]. Additionally, a study leveraging functional near-infrared spectroscopy (fNIRS) from the AI4Pain Grand Challenge dataset introduced Empirically Transformed Energy Patterns and used an optimized ensemble classifier to achieve 91.41% accuracy for binary pain classification and 68.20% for multiclass classification, demonstrating the effectiveness of brain-based sensing combined with machine learning for objective pain assessment [12].

2.2 Multimodal Approaches

Recent works have emphasized the benefits of fusing multiple physiological signals for more robust and generalizable pain classification. A recent review addressed the ongoing challenges in pain assessment, emphasizing its subjective and multifaceted nature, which often necessitates the integration of multiple sensor modalities [13]. The review specifically focused on acute pain assessment and examined three core areas: (a) the types and combinations of sensor modalities used, (b) techniques for fusing data from heterogeneous sensors, and (c) the application of artificial intelligence (AI) methods to multimodal sensor data for pain classification. The authors conducted a comprehensive literature search across IEEE Xplore, Scopus, and Google Scholar, targeting studies published between January 2015 and September 2024. A total of 31 relevant studies were analyzed. The findings highlight a growing trend in leveraging physiological signals particularly from cardiac, electrodermal, and neural sources combined with domain knowledge and deep learning approaches to improve the reliability and accuracy of pain recognition systems. The review also outlines key challenges and proposes future directions for developing more robust and generalizable AI-based pain assessment frameworks.

2.3 Machine Learning for Pain Classification

Various machine learning models have been explored for pain level prediction, including random forests, support vector machines, XGBoost, and more recently, deep neural networks (DNNs) and convolutional neural networks (CNNs). Some methods employ end-to-end deep learning pipelines, while others rely on window-based feature extraction followed by classical classifiers. Despite their promising results, many of these models lack interpretability, which limits their applicability in clinical settings.

While promising, existing works face several challenges: (i) limited generalization across subjects due to inter-individual variability, (ii) insufficient focus on real-time applicability, and (iii) lack of interpretability and explainability. Furthermore, datasets with clearly labeled pain levels and diverse physiological signals are scarce, making benchmarking difficult.

In contrast to earlier works, our study leverages a subject-wise multimodal dataset with structured pain stimuli and proposes a deep learning approach with explainable outputs. We incorporate HRV features derived from BVP signals, integrate windowed temporal context, and emphasize model interpretability to enhance clinical reliability. Our contributions address both accuracy and explainability, aligning with the growing need for trustworthy AI in healthcare.

3 Methodology

Our model development follows a typical machine learning pipeline involving pre-processing, feature extraction, and training phases. The pre-processing involves signal filtering, noise and artifact removal, and subject and trial-wise data separation. The details of the processing steps are described in the following sections.

3.1 Dataset Description

In this work, we utilized the AI4Pain Challenge dataset, which is a controlled environmental study conducted at the Human-Machine Interface Laboratory, University of Canberra, Australia. The dataset focuses on the detection of pain and physiological affect, distinguishing between different levels of pain, such as no pain, low pain, and high pain. It contains different physiological signals such as electrodermal activity (EDA), blood volume pulse (BVP), respiration (RESP), and peripheral oxygen saturation (SpO_2), collected using Biosignal Plux sensors. All physiological data were sampled at 100 Hz. The data were collected from 65 participants (23 female, 42 male), with a mean age of 29.06 and a standard deviation of 8.28, each contributing data for approximately 300 seconds excluding the rest period. During the experiment, pain was induced using electrodes from a transcutaneous electrical nerve stimulation (TENS) machine placed on the inner forearm and on the back of the hand of the participant. For the challenge purpose, the dataset is divided into training (41 participants), validation (12 participants), and testing (12 participants) sets.

3.2 Data Pre-processing

In our preprocessing pipeline, we performed signal cleaning and artifact correction as a first step. For each modality, we applied signal-specific denoising methods such as bandpass filtering and peak correction for BVP, and convex optimization-based decomposition for EDA to reduce noise and remove physiological artifacts. Segments with clearly missing or flatline signals were excluded from feature extraction.

Specifically, to preprocess the electrodermal activity (EDA) signal, we first applied a 4th-order low-pass Butterworth filter with a cutoff frequency of 5 Hz to eliminate high-frequency noise. The filtered signal was then further cleaned using the NeuroKit2 [16] library to smooth the waveform and remove residual artifacts. Based on this cleaned signal, we extracted both tonic and phasic components, following standard procedures reported in the literature [22, 24]. The tonic component represents slow-varying baseline activity, while the phasic component captures rapid, stimulus-evoked responses. Accordingly, both components were utilized

in our analysis to investigate their respective correlations with pain levels. In addition, the original EDA signal was retained, as it contains valuable physiological information [6].

To extract cardiovascular information aligned with normal heart rate (60–100 bpm), the BVP data were filtered using a Butterworth band-pass filter with a frequency range of 0.04–1.7 Hz [5, 14]. Similarly, respiration signals were filtered using a Butterworth band-pass filter with a frequency range of 0.1–0.5 Hz to isolate typical respiratory patterns while reducing high-frequency noise and low-frequency drift [5, 20]. Finally, the peripheral oxygen saturation (SpO_2) signal was filtered using a 3rd order low-pass Butterworth filter with a cutoff frequency of 0.4 Hz to attenuate high-frequency noise [9]. The resulting signal was clipped to a physiologically valid range of 70–100% to ensure consistency and reliability in subsequent analysis.

3.3 Feature Extraction

In order to extract features from the precess data, we segmented the physiological signals into 10-second non-overlapping segments, suggested in the literature [5]. For feature extraction, the preprocessed signals were segmented using a fixed-length windowing strategy. A sampling rate of 100 Hz and a window size of 10 seconds (1,000 samples) were used. For each subject and condition combination, features were extracted differently based on the label type. For the baseline condition (no pain), 6 non-overlapping 10-second windows (totaling 60 seconds) were used to capture variability across the no pain condition. For the low pain and high pain conditions, only the first 10-second window of each trial was considered, ensuring a consistent and comparable input duration across pain-inducing conditions.

In this paper, the feature selection was strictly conducted using only the training and validation data, as the test set was provided without labels by the challenge organizers. Therefore, our identification of the best-performing features was based solely on cross-validation performance within the train and validation split. These selected features were then applied to the test set without any further tuning or re-selection, ensuring that no information from the test set influenced the model.

This approach aligns with standard machine learning protocols and challenge rules, maintaining the integrity of the evaluation and avoiding data leakage. Our goal was to ensure reproducibility and practical relevance by selecting features that consistently generalized well across training and validation sets before deployment on the unseen test set.

3.3.1 Electrodermal Activity (EDA) Feature Extraction. For electrodermal activity (EDA) feature extraction, we used the raw EDA signal along with its tonic and phasic components, processed within a 10 second window. From the raw EDA, we extracted statistical and temporal features including the mean (MSC), standard deviation (SDSC), local segmental variability (MSDSC), the maximum first-order difference (SC-MaxD), and rate of change over 5 and 10-second intervals (SC-Rate5, SC-Rate10). Skin conductance responses (SCRs) were detected from the phasic component using a peak detection algorithm with a minimum height of 0.01 μS and a minimum distance of 0.5 seconds. Based on these peaks, we computed the SCR frequency (CDA.nSCRs), average and total amplitude (CDA.SCR, CDA.AmpSum), and integrated SCR (CDA.ISCR). Similar features were computed using the tonic-to-phasic transformation (TTP), including TTP.nSCRs and TTP.AmpSum. From the tonic signal, we extracted features such as the mean (MSCL), standard deviation (SDSCL), segmental variability (MSDSCL), and rate of change (SCL-Rate5, SCL-Rate10).

Table 1: Description of the extracted Electrodermal Activity (EDA) features.

EDA Feature	Short Description
MSC	Mean of the raw EDA signal
SDSC	Standard deviation of the raw EDA signal
MSDSC	Mean of short-term local standard deviations of EDA
SC_MaxD	Maximum difference between consecutive EDA samples
SC_Rate5	Change in mean EDA over first and second 1-second windows
SC_Rate10	Change in mean EDA over first and last 1-second windows
CDA.nSCRs	Number of skin conductance responses (SCRs) per minute
CDA.AmpSum	Sum of SCR amplitudes
CDA.SCR	Mean amplitude of SCRs
CDA.ISCR	Integrated SCR, area under the phasic curve
TTP.nSCRs	Number of SCRs per minute (duplicate for compatibility)
TTP.AmpSum	Total amplitude of all detected SCRs
MSCL	Mean of the tonic component (SCL)
SDSCL	Standard deviation of the tonic component
MSDSCL	Mean of local short-term std deviations of tonic signal
SCL_Rate5	Change in mean tonic level over first and second 1-second windows
SCL_Rate10	Change in mean tonic level over first and last 1-second windows
Phasic_max	Maximum value of the phasic component
Phasic_mean	Mean of the phasic signal
Phasic_std	Standard deviation of the phasic signal
Phasic_min	Minimum value of the phasic signal
Phasic_median	Median of the phasic signal
Phasic_range	Range of the phasic signal (max - min)
Phasic_iqr	Interquartile range of the phasic signal
Tonic_d1_mean	Mean of the first derivative of tonic signal
Tonic_d1_std	Standard deviation of the first derivative of tonic
Tonic_d2_mean	Mean of the second derivative of tonic signal
Tonic_d2_std	Standard deviation of the second derivative of tonic
wavelet_0.78-1.56Hz_mean	Mean of wavelet coefficients (scale 3, 0.78–1.56 Hz)
wavelet_0.78-1.56Hz_median	Median of wavelet coefficients at same scale
wavelet_0.78-1.56Hz_std	Standard deviation of those coefficients
wavelet_0.39-0.78Hz_mean	Mean of wavelet coefficients (scale 2, 0.39–0.78 Hz)
wavelet_0.39-0.78Hz_median	Median of wavelet coefficients at same scale
wavelet_0.39-0.78Hz_std	Standard deviation of those coefficients
wavelet_0.20-0.39Hz_mean	Mean of wavelet coefficients (scale 1, 0.20–0.39 Hz)
wavelet_0.20-0.39Hz_median	Median of wavelet coefficients at same scale
wavelet_0.20-0.39Hz_std	Standard deviation of those coefficients

Additional statistics were computed from the phasic signal, including maximum, mean, standard deviation, minimum, median, range, and interquartile range. To capture temporal dynamics of the tonic signal, we also calculated its first and second-order derivatives (mean and standard deviation of each). Finally, we applied discrete wavelet decomposition (Daubechies-1, level 8) to the tonic component and extracted the mean, median, and standard deviation of detail coefficients at three frequency bands (0.20-0.39 Hz, 0.39–0.78 Hz, and 0.78–1.56 Hz). Table 1 lists all the features extracted from the EDA data.

3.3.2 Feature Extraction from Blood Volume Pulse (BVP) Data. From each 10-second window of the filtered BVP signal, we extracted a comprehensive set of features spanning multiple domains: time, frequency, time–frequency, and heart rate variability (HRV). Time-domain features include statistical measures such as the mean (overall average signal amplitude), standard deviation (variability), energy (signal power), arc length (signal complexity), skewness (asymmetry), kurtosis (peakedness), and entropy (irregularity). First- and

Table 2: Description of the extracted Blood Volume Pulse (BVP) features.

BVP Feature	Short Description
Mean	Mean of the filtered BVP signal
STD	Standard deviation of the signal
Energy	Signal energy calculated as the sum of squares
ArcLen	Arc length, sum of absolute first differences
Skew	Skewness of the signal distribution
Kurt	Kurtosis of the signal distribution
Entropy	Shannon entropy of the histogram of signal values
MFD	Mean of the first derivative (diff1)
FDD_STD	Standard deviation of the first derivative
FDMN	Ratio of mean diff1 to std of signal
SDD	Mean of the second derivative (diff2)
SDD_STD	Standard deviation of the second derivative
SDDMN	Ratio of mean diff2 to std of signal
HF_PSD	Power in the high-frequency band (0.15–0.4 Hz)
Rel_HF_PSD	Relative power of high-frequency band (HF/total power)
LF_PSD	Power in the low-frequency band (0.04–0.15 Hz)
Rel_LF_PSD	Relative power of low-frequency band (LF/total power)
VLF_PSD	Power in the very low-frequency band (0.0033–0.04 Hz)
Rel_VLF_PSD	Relative power of very low-frequency band (VLF/total power)
STFT_mean	Mean of short-time Fourier transform magnitudes
STFT_std	Standard deviation of STFT magnitudes
STFT_var	Variance of STFT magnitudes
STFT_skew	Skewness of STFT magnitudes
STFT_kurtosis	Kurtosis of STFT magnitudes
HRV_MeanNN	Mean inter-beat (RR) interval in ms
HRV_SDNN	Standard deviation of RR intervals
HRV_RMSSD	Root mean square of successive RR differences
HRV_pNN50	Percentage of RR differences > 50 ms
HRV_MinNN	Minimum RR interval
HRV_MaxNN	Maximum RR interval
HRV_MedianNN	Median RR interval
HRV_IBI_count	Total number of inter-beat intervals
HRV_SD1	Short-term HRV from Poincaré plot
HRV_SD2	Long-term HRV from Poincaré plot
HRV_SD2SD1	Ratio of SD2 to SD1
HRM	Mean heart rate (beats per minute)
STDHR	Standard deviation of heart rate

second-order derivatives of the signal were used to compute features like mean and standard deviation of the first derivative (MFD, FDD_STD), and second derivative (SDD, SDD_STD), as well as normalized versions such as FDMN and SDDMN, which quantify the rate and smoothness of signal changes.

The frequency domain features were calculated using the Welch method. These include absolute and relative power in the very-low-frequency (VLF: 0.0033–0.04 Hz), low-frequency (LF: 0.04–0.15 Hz), and high-frequency (HF: 0.15–0.4 Hz) bands. VLF power is often associated with thermoregulatory or hormonal factors, LF power reflects both sympathetic and parasympathetic nervous activity, and HF power is primarily linked to parasympathetic (vagal) activity. Relative powers help normalize the influence of total power variability across individuals.

For heart rate variability (HRV) analysis, we identified systolic peaks in the BVP signal and calculated inter-beat intervals (IBIs). From these IBIs, we derived standard HRV metrics: MeanNN (average IBI), SDNN (standard deviation of IBI), RMSSD (root mean square of successive IBI differences), and pNN50 (percentage of adjacent IBIs differing by more than 50 ms), which represent beat-to-beat variability and parasympathetic activity. Additional features include minimum, maximum, and median IBIs, total number of beats (IBI_count), as well as SD1 and SD2. SD1 indicates short-term variability, SD2 reflects both short- and long-term variability, and their ratio (SD2/SD1) gives insights into the balance between these two components. Lastly, HRM and STDHR represent the mean and variability of heart rate over the window. Table 2 lists all the features extracted from the BVP data.

3.3.3 Feature Extraction from Respiration Data. From each 10-second window of respiration signal we extracted amplitude based features included the minimum (RESP_Amplitude_Min), maximum (RESP-Amplitude-Max), and mean (RESP-Amplitude-Mean) amplitude between paired peaks and troughs, reflecting the strength and depth of respiration cycles. Basic statistical features such as mean (RSP_Mean), standard deviation (RSP_STD), variance (RSP_Var), median (RSP_Median), minimum (RSP_Min), maximum (RSP_Max), interquartile range (RSP_IQR), range (RSP_Range), coefficient of variation (RSP_CV), and mode (RSP_Mode) were computed to summarize the distribution and variability of the respiration signal.

To characterize signal shape and complexity, we extracted features such as skewness (RSP-Skew), kurtosis (RSP-Kurtosis), Shannon entropy (RSP-Entropy), total signal energy (RSP_Energy), zero-crossing count (RSP-ZeroCrossings), arc length (RSP-ArcLen), and area under the absolute curve using the trapezoidal rule (RSP-Area-Under-Curve). These features provide insight into the irregularity, oscillatory nature, and complexity of the respiratory waveform. Additionally, derivative-based features were computed from the first and second differences of the signal. These include the mean and standard deviation of the first derivative (RSP-Diff1-Mean, RSP-Diff1-STD) and second derivative (RSP-Diff2-Mean, RSP-Diff2-STD), capturing the rate of change and acceleration in respiration dynamics. To capture the morphological structure of the waveform, we also counted the number of local maxima and minima (RSP-NumPeaks and RSP-NumTroughs), representing the number of complete respiratory cycles within each window. Table 3 lists all the features extracted from the respiration signal.

3.3.4 Feature Extraction from Peripheral Oxygen Saturation (SpO_2) Data. From each 10-second window of the cleaned SpO_2 signal a comprehensive set of features was extracted to capture its statistical, morphological, and temporal dynamics. Basic statistical features included the mean, standard deviation (STD), variance (Var), median, minimum, maximum, interquartile range (IQR), range, and coefficient of variation (CV), which together describe the central tendency and dispersion of the signal. The mode represented the most frequent SpO_2 value observed within the window.

Higher-order statistical descriptors such as skewness and kurtosis captured the asymmetry and peakedness of the SpO_2 distribution, respectively, while Shannon entropy quantified the signal's unpredictability or complexity. Morphological and shape-based features included the signal energy (sum of squared amplitudes), arc length (total absolute change in amplitude), zero-crossings (number of times the signal crossed the zero level, indicating variability), and area under the curve (AUC) computed using trapezoidal integration of the absolute signal values.

Additionally, dynamic features were computed from the first and second derivatives of the signal to characterize changes over time. These included the mean and standard deviation of the first

Table 3: Description of the extracted Respiration (RESP) features.

RESP Feature	Short Description
RESP_Amplitude_Min	Minimum amplitude between paired peaks and troughs
RESP_Amplitude_Max	Maximum amplitude between paired peaks and troughs
RESP_Amplitude_Mean	Mean amplitude between paired peaks and troughs
RSP_Mean	Mean of the filtered respiration signal
RSP_STD	Standard deviation of the signal
RSP_Var	Variance of the signal
RSP_Skew	Skewness of the signal distribution
RSP_Kurtosis	Kurtosis of the signal distribution
RSP_Entropy	Shannon entropy of the histogram of signal values
RSP_Energy	Signal energy calculated as the sum of squares
RSP_ZeroCrossings	Number of times the signal crosses the zero line
RSP_ArcLen	Arc length, or cumulative absolute first differences
RSP_Area_Under_Curve	Area under the absolute signal curve (trapezoidal integration)
RSP_Diff1_Mean	Mean of the first derivative of the signal
RSP_Diff1_STD	Standard deviation of the first derivative
RSP_Diff2_Mean	Mean of the second derivative of the signal
RSP_Diff2_STD	Standard deviation of the second derivative
RSP_NumPeaks	Number of detected peaks (inhalation events)
RSP_NumTroughs	Number of detected troughs (exhalation events)
RSP_Median	Median of the signal
RSP_Min	Minimum signal value
RSP_Max	Maximum signal value
RSP_IQR	Interquartile range (75th - 25th percentile)
RSP_Range	Range (max - min) of the signal
RSP_CV	Coefficient of variation (std/mean)
RSP_Mode	Mode of the signal (most frequent value)

Table 4: Description of the extracted peripheral Oxygen Saturation (SpO_2) features.

SpO_2 Feature	Short Description
SpO_2_Mean	Mean of the filtered SpO_2 signal
SpO_2_STD	Standard deviation of the SpO_2 signal
SpO_2_Var	Variance of the SpO_2 signal
SpO_2_Min	Minimum SpO_2 value in the window
SpO_2_Max	Maximum SpO_2 value in the window
SpO_2_Median	Median of the SpO_2 signal
SpO_2_IQR	Interquartile range of the SpO_2 signal
SpO_2_Range	Difference between maximum and minimum values
SpO_2_CV	Coefficient of variation (std/mean)
SpO_2_Mode	Most frequent SpO_2 value in the signal
SpO_2_Skew	Skewness of the SpO_2 signal distribution
SpO_2_Kurtosis	Kurtosis of the SpO_2 signal distribution
SpO_2_Entropy	Shannon entropy of signal histogram
SpO_2_Energy	Total signal energy (sum of squares)
SpO_2_ZeroCrossings	Number of times signal crosses zero
SpO_2_ArcLen	Arc length, sum of absolute first differences
SpO_2_Area_Under_Curve	Trapezoidal area under absolute signal curve
SpO_2_Diff1_Mean	Mean of the first derivative of the signal
SpO_2_Diff1_STD	Standard deviation of the first derivative
SpO_2_Diff2_Mean	Mean of the second derivative of the signal
SpO_2_Diff2_STD	Standard deviation of the second derivative

derivative (Diff1-Mean, Diff1-STD), which reflect the rate of change, and the mean and standard deviation of the second derivative (Diff2-Mean, Diff2-STD), which capture the acceleration or curvature of the signal trajectory. Table 4 lists all the features extracted from the peripheral oxygen saturation data.

3.4 Classification Models

In this paper, we conducted a series of experiments to explore which combination of physiological signal modalities provides the best classification performance to identify pain levels. Figure 1 illustrates the overview of our classification process. We treated the baseline recordings as the No Pain condition and excluded the post-stimulus rest periods, as these may retain residual pain effects. To classify the three pain conditions (No Pain, Low Pain, and High Pain) we evaluated several traditional machine learning models, including Random Forest (RF), Gradient Boosting (GB), Support Vector Machine (SVM), and XGBoost (XGB). Among these, the XGB model consistently outperformed the others in terms of accuracy and generalizability.

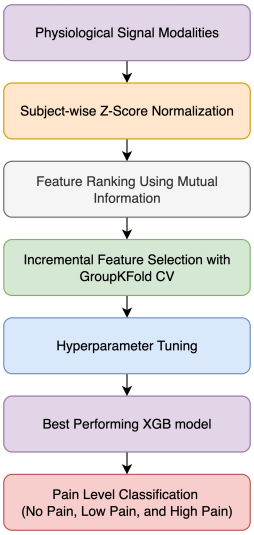


Figure 1: An overview of the classification process using GroupKFold cross validation for performance evaluation.

We developed a subject-independent pipeline using XGB, where we first applied z-score normalization individually for each subject to mitigate inter-subject physiological variability. We then ranked all features using mutual information scores to assess their relevance to the pain level labels. Based on these rankings, we incrementally selected subsets of top features and evaluated their performance using a 5-fold GroupKFold cross-validation strategy that ensured no subject appeared in multiple fold. This approach allowed for unbiased and robust model evaluation. Once the optimal feature subset was identified, we fine-tuned the XGB model's hyperparameters such as the number of estimators, learning rate, tree depth, and sampling rates through an exhaustive grid search. The final model was trained using the combined training and validation sets, and was subsequently used to predict pain level labels on an independent, unseen test set.

3.5 Interpretation of Best Model

To make our model interpretable, we evaluated Permutation Feature Importance, which is a model-agnostic technique used to interpret machine learning models by evaluating the impact of each feature on prediction performance. It works by randomly shuffling the values of a single feature and measuring the resulting decrease in model accuracy. A significant drop indicates the feature's strong contribution to the model's decision-making. This method provides

an intuitive and scalable way to understand which physiological signal features most influence pain classification, enhancing the interpretability of our predictive framework.

4 Results and Discussion

In this challenge, the final test dataset evaluation results were evaluated using the accuracy metric. In addition to accuracy, we evaluate our models based on other metrics such as precision, recall, and F1-score on the validation dataset (Table 5). For the test data, since the pain labels were not available, we report only the test results provided by the organizers after submitting the predicted labels.

The performance of the modalities on both the validation and test sets, along with their comparison to the baseline results, is presented in Table 6. The performance comparison across different sensor modality combinations reveals that combining multiple physiological signals can enhance model accuracy, particularly on the validation set. Using electrodermal activity (EDA) alone yielded a validation accuracy of 73.0% but a comparatively lower test accuracy of 57.76%. In contrast, blood volume pulse (BVP) alone achieved a slightly lower validation accuracy (68.0%) but the highest test accuracy (62.93%), indicating better generalization. The fusion of EDA and BVP resulted in the best validation accuracy (80.0%), though the corresponding test accuracy (60.92%) did not surpass that of BVP alone. Similarly, combining EDA, BVP, and respiration (RESP) achieved 80.0% validation accuracy and improved test accuracy to 62.07%. Interestingly, adding all modalities did not further boost performance, as the test accuracy remained at 60.92%. These findings suggest that while combining signals improves model learning, excessive modality fusion does not always guarantee superior generalization, and BVP alone may offer the most stable predictive performance on unseen data.

Table 5: Class-wise Precision, Recall, and F1-score across sensor modalities in validation dataset.

Sensor Modality	Precision			Recall			F1-score		
	No-Pain	Low	High	No-Pain	Low	High	No-Pain	Low	High
EDA	0.78	0.73	0.70	0.67	0.70	0.78	0.72	0.72	0.74
BVP	0.73	0.64	0.70	0.68	0.62	0.74	0.71	0.63	0.72
EDA + BVP	0.83	0.78	0.80	0.72	0.77	0.85	0.77	0.78	0.83
EDA + BVP + RESP	0.88	0.81	0.77	0.75	0.76	0.87	0.81	0.78	0.81
All Modalities	0.84	0.76	0.80	0.70	0.76	0.85	0.76	0.76	0.83

Table 6: Model performance on validation and test sets across different sensor modalities.

Sensor Modality	Validation Accuracy (%)	Test Accuracy (%)	Val. Baseline Acc. (%)	Test Baseline Acc. (%)
EDA	73.0	57.76	45.60	38.50
BVP	68.0	**62.93**	56.25	45.69
EDA + BVP	**80.0**	60.92	-	-
EDA + BVP + RESP	**80.0**	**62.07**	-	-
All Modalities	79.0	60.92	52.77	52.30

Figure 2 illustrates a comparative analysis of validation accuracies across four physiological sensor modalities, EDA, BVP, RESP, and SpO_2 against their respective baseline performances. Among these, EDA achieved the highest validation accuracy at 73.0%, representing a substantial improvement of 27.4% over the baseline (45.60%). BVP also performed well, improving from 56.25% to 68.0%. Moderate gains were observed in RESP (49.07% to 55.0%) and SpO_2 (37.26% to 53.0%).

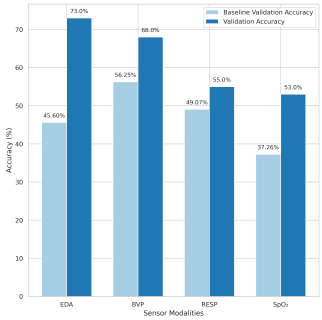


Figure 2: Comparison of validation accuracy across different sensor modalities, including baseline performance.

The results suggest that EDA and BVP are the most informative modalities for short-term pain classification, as their individual and combined use consistently yielded the highest validation and test accuracies. In contrast, the inclusion of respiration (RESP) and peripheral oxygen saturation did not significantly improve performance. This is likely because RESP and SpO_2 signals reflect slower physiological responses that may not change rapidly or distinctly enough during short-duration pain stimuli. While these modalities may be useful for monitoring long-term stress or chronic conditions, their limited responsiveness to brief pain events makes them less effective in this context. Therefore, EDA and BVP appear to be more sensitive and reliable for capturing the rapid autonomic changes associated with acute pain.

Figure 3, 4 and 5 present confusion matrices and ROC curves for the classification of pain levels using BVP, EDA and combined EDA+BVP features using validation set. The combined feature model achieved high classification accuracy across all pain levels, particularly for High Pain (AUC = 0.92) and No Pain (AUC = 0.96), demonstrating strong discriminative ability.

The EDA-only model also performed well, especially in detecting No Pain (AUC = 0.92), but showed moderate confusion between High and Low Pain. Similarly, the BVP-only model yielded competitive results with strong detection of No Pain (AUC = 0.94), though it struggled more in differentiating Low and High Pain levels compared to the multimodal setup.

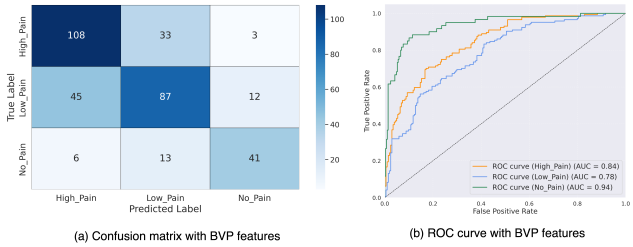
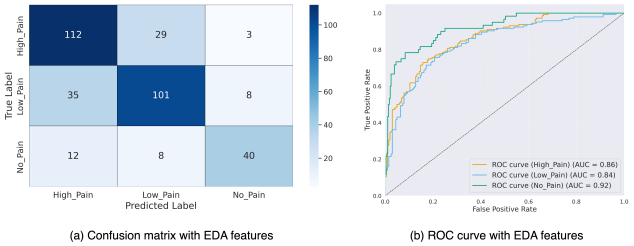


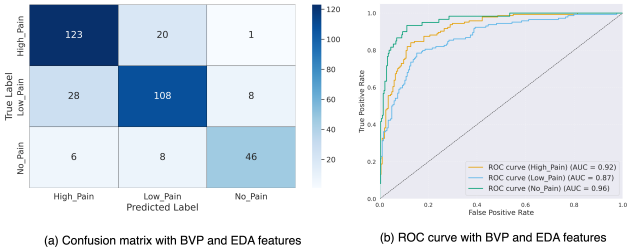
Figure 3: Confusion matrix and ROC curve using BVP features evaluated on the validation dataset.

These results highlight the effectiveness of multimodal signal fusion to improve pain classification performance, particularly when distinguishing subtle variations between Low and High Pain states.



(a) Confusion matrix with EDA features (b) ROC curve with EDA features

Figure 4: Confusion matrix and ROC curve using EDA features evaluated on the validation dataset.



(a) Confusion matrix with BVP and EDA features (b) ROC curve with BVP and EDA features

Figure 5: Confusion matrix and ROC curve using BVP and EDA features evaluated on the validation dataset.

However, considering the current clinical reality, where assessments still rely heavily on subjective scales like NRS (Numeric Rating Scale) or VAS (Visual Analog Scale) and facial expressions, there is certainly a demand for more reproducible and objective measures. So rather than evaluating the sufficiency based solely on accuracy, we believe the key lies in the implementation strategy in clinical practice, especially in terms of user interface and experience. We feel the real challenge is in designing a proper human-in-the-loop system. Even if we had 99% accuracy, without a well-thought-out integration with clinical workflow, it would not be truly useful.

In that sense, we would say that our achieved accuracy might be the minimum viable level required to design a sufficiently effective UI/UX for clinical implementation.

Figure 6, and 7 illustrate permulation features importance while using EDA and BVP features. For BVP, HRV complexity and temporal dynamics dominate in feature importance. ArcLen and SDD_STD geometric and statistical measures of heart rate complexity. HRV MeanNN, HRM, HRV_MaxNN highlight the importance of interbeat intervals and heart rate measures. Pain related variations in BVP are best captured using a mix of time-domain, frequency-domain, and non-linear HRV metrics, with a strong emphasis on complexity and variability.

For EDA in Figure 7, shows the most important features are derived from tonic/phasic decomposition and wavelet-based analysis. These results confirm that interpretable and domain-specific features significantly enhance model performance and understanding in multimodal pain detection. Unlike black-box deep learning models, our XGBoost-based approach provides interpretability through

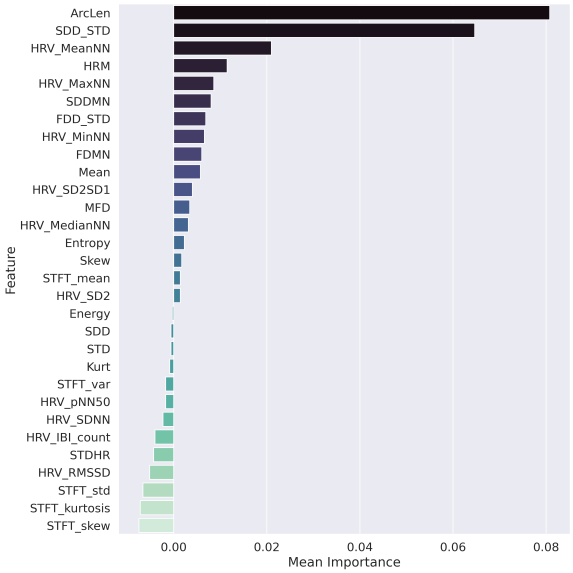


Figure 6: Permutation feature importance for the best model-BVP

feature importance analysis, offering insights into which physiological signals most influence pain prediction, an essential aspect for clinical trust.

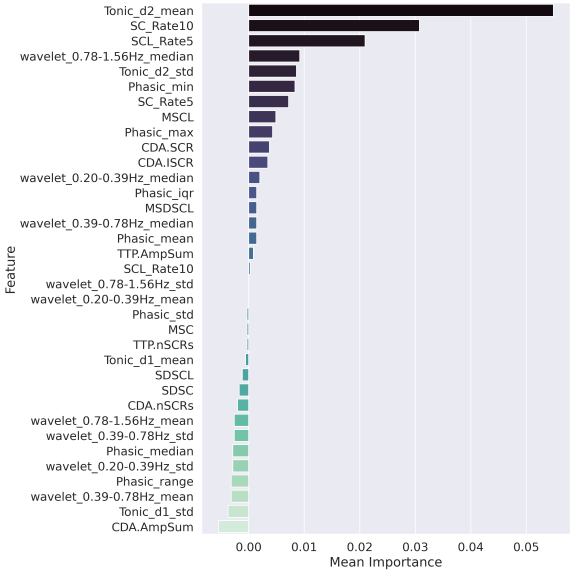


Figure 7: Permutation feature importance for the best model-EDA

5 Conclusion

In this study, we used the AI4Pain challenge dataset to understand human physical pain based on physiological signals such as EDA, BVP, RESP, and SpO_2. We extracted both general and signal specific features from each modality and applied a feature selection method

to identify the most important features for detecting different pain levels. Our findings demonstrate promising results for classifying pain levels (No Pain, Low Pain, and High Pain) using a multimodal approach. Specifically, the combination of EDA and BVP features, trained with a fine-tuned XGBoost model, achieved a validation accuracy of 80% and a test accuracy of 60.92%. Notably, when using only the BVP signal and incorporating both statistical and heart rate variability (HRV) features, the model attained the highest test accuracy of 62.93%.

Although the challenge encourages a multimodal approach, our experiments indicate that incorporating additional modalities such as RESP and SpO_2 did not improve the performance of the model. In fact, the inclusion of these signals resulted in a slight decline in accuracy. Furthermore, we compared the performance of each sensor modality against the baseline results provided by the organizers, and in all cases, our models consistently outperformed the baseline benchmarks.

Acknowledgments

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